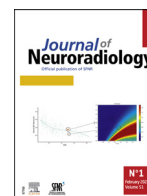




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Review

MRI features and disability in multiple sclerosis: A systematic review and meta-analysis



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ABSTRACT

Background: In this systematic review and meta-analysis, we aimed to investigate the correlation between disability in patients with Multiple sclerosis (MS) measured by the Expanded Disability Status Scale (EDSS) and brain Magnetic Resonance Imaging (MRI) features to provide reliable results on which characteristics in the MRI can predict disability and prognosis of the disease.

Methods: A systematic literature search was performed using three databases including PubMed, Scopus, and Web of Science. The selected peer-reviewed studies must report a correlation between EDSS scores and MRI features. The correlation coefficients of included studies were converted to the Fisher's z scale, and the results were pooled.

Results: Overall, 105 studies A total of 16,613 patients with MS entered our study. We found no significant correlation between total brain volume and EDSS assessment (95 % CI: -0.37 to 0.08; z-score: -0.15). We examined the potential correlation between the volume of T1 and T2 lesions and the level of disability. A positive significant correlation was found (95 % CI: 0.19 to 0.43; z-score: 0.31), (95 % CI: 0.17 to 0.33; z-score: 0.25). We observed a significant correlation between white matter volume and EDSS score in patients with MS (95 % CI: -0.37 to -0.03; z-score: -0.21). Moreover, there was a significant negative correlation between gray matter volume and disability (95 % CI: -0.025 to -0.07; z-score: -0.16).

Conclusion: In conclusion, this systematic review and meta-analysis revealed that disability in patients with MS is linked to extensive changes in different brain regions, encompassing gray and white matter, as well as T1 and T2 weighted MRI lesions.

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Introduction

Multiple sclerosis (MS) which is characterized by neurological disabilities is an autoimmune disease with an increasing incidence, especially among young people.^{1–3} Patients with MS can show a wide variety of neurological symptoms such as vision loss, double vision (diplopia), gait impairment, bladder dysfunction, and numbness.^{4,5} The pathology of MS is described as demyelination, neurodegeneration, and proliferation of astroglia in the central nervous system (CNS).⁶ Due to the type of disease process, MS can be divided into four subgroups including clinically isolated syndrome (CIS), relapsing-remitting MS (RRMS), primary progressive MS (PPMS), and secondary progressive MS (SPMS).⁷ Some studies have suggested the aforementioned prognostic factors reflecting poor prognosis of patients with MS: male gender, non-Caucasian races,

progressive type of disease from the onset, and cognitive impairment.⁸ Moreover, MS is a high-burden disease with rising healthcare costs which also, reduces the quality of life (QoL) of MS patients and in turn can cause psychological problems.⁹ Magnetic Resonance Imaging (MRI), cerebrospinal fluid (CSF) analysis through which 2 or more oligoclonal bands (OCBs) can be detected in MS patients, and evoked potential testing which can reflect demyelination features such as reduced conduction speed in central motor alongside physical examination and history taking are considered as most reliable diagnostic tools in MS disease.^{10,11}

Among the mentioned diagnostic tools MRI is the most sensitive and reliable technique in the detection of MS patients since it can evaluate different parameters of MS disease including lesion load, disease activity, brain volume, and changes of white and gray matter.^{4,10,12} The McDonald criteria considering the location of lesions in CNS are introduced to facilitate the diagnosis of MS patients. In this way, new MRI features such as conventional T2-weighted and post-contrast T1-weighted MRI can improve the

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assessment of McDonald criteria for earlier diagnosis of patients with MS.¹³ Moreover, the introduction of new MRI methods such as diffusion tensor imaging and T1 relaxometry demonstrated more CNS changes in patients with MS.^{14,15} To increase the sensitivity of these criteria, other differential diagnoses should be excluded through primary diagnostic evaluations such as CSF analysis.^{13,16,17} Brain volume loss is a common imaging finding among MS patients at different stages and it has been evaluated as an effective outcome in several studies.^{18–20} Also, deep gray matter (DGM) changes are other MRI findings in MS patients which include damage in structures such as the thalamus, putamen, and globus pallidus.^{5,21–23} Using DTI which is a sensitive MRI technique to assess the white matter lesion in MS, it is shown that compared to healthy people, MS patients reflect decreased fractional anisotropy (FA) and increased mean diffusivity (MD).²⁴ Moreover, reduction in the T1/T2 ratio in normal-appearing white matter (NAWM) regions such as the cingulate bundle, corpus callosum, temporal and parietal NAWM, and atrophies of different brain regions including the occipital, parietal, and frontal cortex were reported in MS patients.²⁵

Expanded Disability Status Scale (EDSS) described by Kurtzke is the most common method in the evaluation of MS severity and functional deficits.²⁶ This assessment has rates of 0 (normal neurological status) to 10 (MS-related death) rates with an interval of 0.5. EDSS is used in the evaluation of MS progression and treatment outcomes in MS patients, for instance between the rates of 4 and 6 it is related to movement disabilities.²⁷ Previous studies have reviewed the correlation between MRI findings and EDSS scores in MS patients. In a meta-analysis done by Valizadeh et al. a positive correlation between T1 hypointense lesions' mean volume and EDSS score was reported in MS patients.²⁸ In another study, Al-Radaideh et al. demonstrated that there were no significant correlations between most DGM metrics and lesions' metrics with MS severity.²⁹ In a study by Elliott et al. it is showed that white matter lesions can play an important role in the prediction of MS progression.³⁰ While Tsagkas et al. in their study have reported a strong correlation between spinal cord volume (SCV) loss and the severity of EDSS score in MS patients.³¹ However, to the best of our knowledge, none of these studies has evaluated several MRI features and EDSS scores.

Due to the mentioned disabilities that MS may bring with it, more MRI biomarkers should be considered in the diagnosis process of MS to increase the sensitivity of this technique and improve the prognosis of MS patients through early detection. In this systematic review and meta-analysis, we aimed to investigate the correlation between disability in patients with MS measured by EDSS and brain MRI features to provide reliable results on which characteristics in the MRI can predict disability and prognosis of the disease.

Materials and methods

The current systematic review and meta-analysis was conducted following the Preferred Reporting for Systematic Review and Meta-Analysis (PRISMA) checklist.³²

Search strategy

A systematic literature search was performed using three databases including PubMed, Scopus, and Web of Science in November 2022. The following terms were used in our search strategy: whole brain, volumetric MR, magnetic resonance imaging, gray matter, white matter, MRI, lesion, hypotense, hyperintense, Multiple sclerosis, disability, progression, EDSS, expanded disability status scale. Additional studies were identified via manual search in the reference list of review articles.

Eligibility criteria

The selected peer-reviewed observational studies including case-controls, cohort, cross-sectional and longitudinal studies must report

a correlation between EDSS score and MRI features including the amygdala, brain parenchymal fraction, caudate volume, cerebellum volume, corpus callosum volume, cortical GM volume, cortical lesion number, cortical lesion volume, cortical T1-lesion volume, cortical thickness, frontal atrophy, frontal volume, globus pallidus, GM fraction, GM volume, hippocampus volume, infratentorial volume, lateral ventricle volume, normal-appearing GM volume, normal-appearing WM volume, nucleus accumbens volume, pons volume, precentral gyrus volume, Putamen volume, subcortical deep GM (SDGM) volume, T-1 lesion volume, T-2 lesion volume, thalamus volume, third ventricle width, Total brain volume, Total lesion volume, WM fraction, WM lesion volume, and WM volume. We excluded review articles, non-English studies, clinical trials, abstracts, case reports, and case series.

Study selection

Two independent investigators (TM, NR) reviewed the title and abstracts in the first step. Then the full text of the selected studies were carefully reviewed and eligible studies were identified. Any disagreements in each step were resolved by consulting with the third investigator (FN).

Data extraction

The same reviewers independently extracted the following data from included studies by consulting with the third researcher (FN): Author, year, country, study design, sample size, mean age, number of females, number of MS patients, type of MS, EDSS score, disease duration, investigated MRI features, and reported correlation between EDSS score and MRI features.

Quality assessments

The quality of the included studies was assessed using the Newcastle–Ottawa scale (NOS) by scoring them in the selection of the participants, comparability of study groups, and outcome assessment.³³

Statistical analysis

To assess statistical heterogeneity and quantify the variability among the included studies, we utilized the I^2 value. I^2 values falling within the ranges of 0–25 %, 25–50 %, 50–75 %, and >75 % were indicative of very low, low, medium, and high heterogeneity, respectively. The I^2 statistic helps to determine the proportion of variation across studies attributed to heterogeneity rather than chance. The I^2 value was computed as follows: $I^2 = 100 \% \times (Q - df)/Q$. The inverse variance method was employed to calculate the weight of each study. Subsequently, the data from all the included studies were combined, and an overall effect size estimate was assessed. The correlation coefficients of the included studies were converted to the Fisher's z scale, and the results were pooled. In cases where $I^2 > 50 \%$, a random-effects model was used due to substantial heterogeneity; otherwise, a fixed-effects model was employed. The R software (version 3.3.3; R Foundation for Statistical Computing, Vienna, Austria) was used for all statistical analyses.

Results

Search results and characteristics of studies

Our literature search yielded 14,692 studies and four articles identified via manual search (Fig. 1). After duplicate removal, 10,537 studies underwent a screening process in two steps. First, 9995 studies were excluded after the title and abstract screening and 542 studies

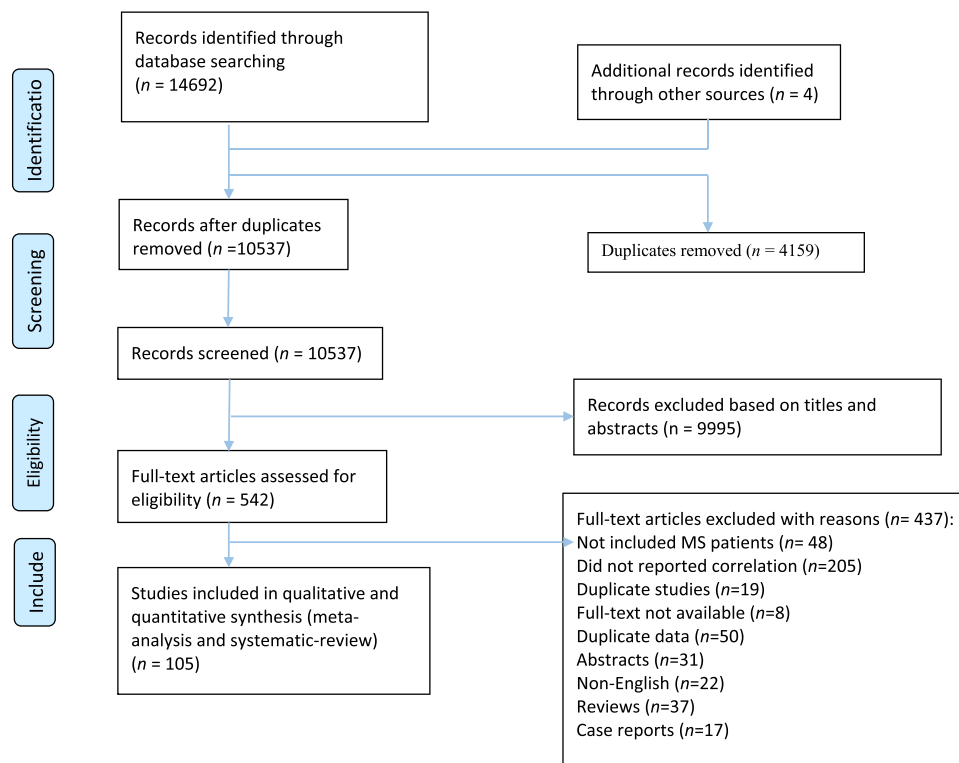


Fig. 1. PRISMA flow diagram depicting the flow of information through the different phases of a systematic review.

remained. Finally, 105 studies were marked eligible to be included in our qualitative and quantitative synthesis after a full-text review.^{34–137}

A total of 16,613 patients with MS entered our study (Table 1). The mean age ranged between 26 and 60. Among included studies 52 were cohorts, 42 were cross-sectional, six were case-control, and three were longitudinal. The average quality score was 6.88 for included studies which can be considered acceptable. The characteristics of the included studies are detailed in Table 1.

Disability and total brain, WM, and GM volumes

Nine studies reported the association between brain volume and EDSS. Our pooled analysis described that there was no significant correlation between total brain volume and EDSS assessment (95 % CI: −0.37 to 0.08; z-score: −0.15) (Figs. 2 and 3). This analysis reflected a considerable heterogeneity among studies ($\chi^2 = 91.14$, $df = 8$, $I^2 = 91$ %). Among these studies, Jacobsen 2014 et al., seem to be an outlier and the main cause of heterogeneity in this analysis.

We observed a significant correlation between WM volume and EDSS score in patients with MS (95 % CI: −0.37 to −0.03; z-score: −0.21) with a moderate heterogeneity ($\chi^2 = 11.40$, $df = 4$, $I^2 = 65$ %). Furthermore, we measured the correlation between WM fraction and disability. There was a significant correlation (95 % CI: −0.25 to −0.01; z-score: −0.13) (Fig. 3). Also, a high level of heterogeneity was observed among these studies ($\chi^2 = 54.45$, $df = 12$, $I^2 = 78$ %). Also, we found a significant correlation between normal-appearing WM volume and EDSS score in patients with MS (95 % CI: −0.30 to −0.04; z-score: −0.17). The heterogeneity of this analysis was not considerable ($\chi^2 = 7.55$, $df = 6$, $I^2 = 21$ %).

Pooling the reported correlation among included studies showed a significant negative correlation between GM volume and disability (95 % CI: −0.025 to −0.07; z-score: −0.16) with no heterogeneity among studies ($\chi^2 = 5.16$, $df = 8$, $I^2 = 0$ %). Moreover, similar to the WM fraction, the GM fraction was significantly associated with the

EDSS score (95 % CI: −0.35 to −0.18; z-score: −0.27). Also, substantial heterogeneity was calculated ($\chi^2 = 48.82$, $df = 16$, $I^2 = 67$ %). However, there was no significant correlation between normal-appearing GM volume and disability (95 % CI: −0.76 to 0.74; z-score: −0.02). There was a high level of heterogeneity among the studies ($\chi^2 = 15.96$, $df = 2$, $I^2 = 87$ %). Similarly, cortical GM volume (95 % CI: −0.18 to 0.20; z-score: 0.01) and subcortical deep gray matter (95 % CI: −0.17 to 0.06; z-score: −0.06) were not associated with EDSS score in patients with MS.

Disability and lesions

We investigated the correlation between total lesion volume and EDSS score. We found a significant correlation between total lesion volume and disability (95 % CI: 0.21 to 0.43; z-score: 0.33) with a low level of heterogeneity ($\chi^2 = 2.94$, $df = 2$, $I^2 = 32$ %). In line with total brain lesion volume, cortical lesion volume was positively associated with disability in MS patients (95 % CI: 0.44 to 0.57; z-score: 0.51). Moderate heterogeneity was observed among studies ($\chi^2 = 6.83$, $df = 4$, $I^2 = 41$ %). However, the number of cortical lesions in contrast to their volume was not associated with the EDSS score (95 % CI: −0.57 to 0.62; z-score: 0.04). The included studies in the analysis were heterogeneous ($\chi^2 = 27.12$, $df = 1$, $I^2 = 96$ %).

We also measured the correlation between T1-Lesion volume and level of disability among patients with MS. A positive correlation was found (95 % CI: 0.19 to 0.43; z-score: 0.31) (Fig. 4) with a high level of heterogeneity ($\chi^2 = 15.55$, $df = 5$, $I^2 = 68$ %). Moreover, T1-Lesion volume in cortical regions was significantly associated with EDSS (95 % CI: 0.45 to 0.84; z-score: 0.69). No heterogeneity was reported between the two included studies ($\chi^2 = 0.41$, $df = 1$, $I^2 = 0$ %).

Another important finding was the significant correlation between T2-Lesion volume and disability (95 % CI: 0.17 to 0.33; z-score: 0.25) with a high degree of heterogeneity ($\chi^2 = 103.06$,

Table 1
Demographic and clinical characteristic of included studies.

Author, year	Country	Type of study	Sample size	Number of MS	mean age of MS	Number of Female	Disease duration	Type of MS	EDSS mean baseline	NOS
Bakshi et al. 2002	USA	Case Control	110	60	MS= 42±9 control=42±10	41	10.6 ± 9.4	RRMS-SPMS	3.7 (0–8)	8
Cocozza et al. 2020	Italy	Case Control	40	40	44.7 ± 12.6	23	8.2 ± 8	RRMS-PMS	2.5 (0–6)	7
Elzayady et al. 2021	Egypt	Case Control	60	30	MS=30.93±3.67 HC=31.13±3.62	37	6.78±3.17	RRMS	2(0.5–5)	7
Varog et al. 2009	Turkey	Case Control	53	27	29±11	27	2.93 ± 2.95	RRMS-SPMS	3.0 ± 1.18	8
Stefano et al. 2007	Switzerland	Case Control	229	229	44.4 ± 11.3	RRMS=132 SPMS=27	12.9 ± 9.1	RRMS-SPMS	3.0 (0–7.5)	8
Bakshi et al. 2008	USA	Case Control	103	103	42.7 ± 9.1	77	14.1 ± 9.2	RRMS-SPMS-PPMS	3.3(0–8.5)	6
D'Hooghe et al. 2019	Belgium	Cohort	470	470	46.41±11.37	318	14.5 ± 9.9	NR	4.2 (0–9)	7
Audoin et al. 2006	France	Cohort	31	21	RRMS= 36 control= 37	RRMS=16 control=4	25.8	RRMS	1(0–3)	7
Audoin et al. 2007	France	Cohort	24	24	31	18	NR	CIS	1 (0–2)	8
Bagnato et al. 2020	USA	Cohort	27	18	MS=45.5 ± 14.6 HC=41.7 ± 10		12.1 ± 10.6	CIS-RRMS-SPMS	1 (0–6.5)	6
Bakshi et al. 2001	USA	Cohort	228	114	43±9	89	9.6 ± 9	RRMS-SPMS	3(0–9)	7
Bakshi et al. 2020	USA	Cohort	200	200	39.5 ± 8.0	148	7.2 ± 6.5	RRMS-SPMS	1.2(0–5.5)	8
Barnett et al. 2021	Australia	Cohort	2875	2875	CIS=37.9 ± 10.6 RRM=42.5 ± 10.6 PMS=53.1 ± 9.7	2156	10.3 ± 8.3	CIS-RRMS-PMS	2.0 (1.5 –3.5)	7
Beer et al. 2016	Germany	Cohort	332	244	MS= 37 ± 10 HC= 31 ± 8	MS= 165 HC= 49		NR	1.29 ± 0.99	7
Bonneville et al. 2002	USA	Cohort	49	49	39(24–55)	36	8.7	RRMS	2(0–6)	5
Brex et al. 2002	UK	Cohort	109	109	32	68	NR	NR	2	7
Daumer et al. 2009	UK	Cohort	578	578	SPMS=44		7.25	RRMS-SPMS	2.5	5
De Stefano et al. 1996	Italy	Cohort	241	206	MS=37 HC=37	MS=146 HC=20	6	RRMS-SPMS-PPMS	2.25(0–8)	8
Dehmeshki et al. 2003	UK	Cohort	60	30	MS=40.7 HC=39.4	25	7.3	PPMS	5.5(2–7)	7
Dupuy et al. 2015	USA	Cohort	45	45	42.3 ± 8.3	33	11.0 ± 7.8	RRMS-SPMS	NR	8
Dwyer et al. 2018	USA	Cohort	192	192	CIS=44.8 ± 11.0 RRMS=43.8 ± 11.1 SP/ PPMS=55.5 ± 7.9	141	12.3 ± 8.4	CIS-RRMS-SPMS-PPMS	2.0(1.5–2.5)	8
Filippi et al. 2001	UK	Cohort	78	78	41.6 ± 9.5	42	10	RRMS-SPMS-PPMS	5.0(0.0–8.5)	6
Fisher et al. 2000	USA	Cohort	172	160	36.1 ± 6.8	77	6.3 ± 5.6		2.34±0.84	7
Fisniku et al. 2008	UK	Cohort	107	107	32	71	20.2	CIS	3	8
Fisniku et al. 2008	UK	Cohort	98	73	MS=51.4 ± 7.2 HC=41.7 ± 7.7	MS=49 HC=14	19.7 ± 1.1	RRMS-SPMS-CIS	2.5(0 – 8)	8
Fisniku et al. 2009	UK	Cohort	88	69	MS=51.4 HC=39.91	MS=45 HC=11		RRMS-SPMS-CIS	2.5(0–8)	8
Furby et al. 2008	UK	Cohort	117	117	50.4 ± 6.9	84	19.9 ± 8.9	SPMS	6.0 (0.5) (IQR)	5
Ge et al. 2000	USA	Cohort	56	36	MS=37.8 HC=33.8	27	3.3	RRMS-SPMS	2.5(1–6.5)	8
Ge et al. 2001	USA	Cohort	45	30	MS=34.9 HC=32.9	MS=24 HC=15	3.8	RRMS	2	8
Giorgio et al. 2014	Italy	Cohort	57	57	33.8 ± 8.1	42	4.9 ± 4.07	RRMS	1.8 ± 1.1	7
Granberg et al. 2015	Sweden	Cohort	46	23	MS=57±8.0 HC=57± 7.2	MS=18 HC=18	27±6.9	RRMS-SPMS	6.0(1.5–8.0)	7
Hardmeier et al. 2003	Germany	Cohort	138	138	38.5 ± 7.1		7.6 ± 5.4	RRMS	3.5(2–5.5)	6
Harrison et al. 2015	USA	Cohort	49	34	MS=43±10 HC=41± 10	MS=18 HC=10	10±7	RRMS-SPMS-PPMS	3.0(1–6.5)	8
Hasan et al. 2011	USA	Cohort	364	109	RRMS=42.6 ± 9.8 SPMS=54.7 ± 9.6 CIS=39.2 ± 14.7	RRMS=68 SPMS=7	RRMS=12.23±11.69 SPMS=22.31±9.7 CIS=2.21±2.74	RRMS-SPMS-CIS	RRMS=1.7(0–6.5) SPMS=4.2(2–6) CIS=0.4(0–2)	6
Hurtado Rúa et al. 2022	USA	Cohort	103	103	41.7 ± 10.9	69	8.5 ± 7.5	RRMS-SPMS	2	8

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Table 1 (Continued)

Author, year	Country	Type of study	Sample size	Number of MS	mean age of MS	Number of Female	Disease duration	Type of MS	EDSS mean baseline	NOS
Iannucci et al. 1999	Italy	Cohort	102	82	MS=34.3 ± 8.3 HC=33.8 ± 5.0	MS=49 HC=12	6	RRMS-SPMS	4.0(1.0–7.0)	7
Jacobsen et al. 2014	Norway	Cohort	67	67	41±9.5	78	101±96 months	RRMS-SPMS-PPMS	3.5(0–6.0)	7
Klistorner et al. 2021	Australia	Cohort	33	33	42.6 ± 10.3	20	6.5 ± 2.8	RRMS	1.1 + 1.0	6
Lapucci et al. 2022	Italy	Cohort	45	45	37.4 ± 6.8 years	30	NR	RRMS-SPMS	3.5 ± 2.0	7
Nazarov et al. 2019	Russia	Cohort	92	92	32.5	60	22 months	NR	2 (0, 6.5)	6
Pirpamer et al. 2022	Hungary	Cohort	166	117	32.6 ± 8.6	79	8	RRMS-CIS	1.5 (1.0–2.0)	8
Rizkallah et al. 2021	Egypt	Cohort	62	31	27.54 ± 4.93	23	NR	RRMS	3.64 ± 1.39	7
Sastre-Garriga et al. 2004	UK	Cohort	88	43	46	19	3.32	PPMS	4.5 (3–7)	6
Sormani et al. 2009	Italy	Cohort	906	906	28±7	70	10±7	RRMS-SPMS	3.5 ± 1.8	6
Vaneckova et al. 2009	Czech Republic	Cohort	181	181	31.2 ± 7.9	142	5.4 ± 5.1	RRMS	1.9 ± 0.9	7
Calabrese et al. 2009	Italy	Cohort	107	107	RRMS=34.2 SPMS=41.1	RRMS=50 SPMS=22	RRMS=4.8 SPMS=11.6	RRMS-SPMS	2.0(1.5–3.5)	7
Calabrese et al. 2011	Italy	Cohort	147	105	MS=33.3 ± 11 HC=33.5 ± 8.6	MS=76 HC=31		CIS	2.1(0–3.5)	5
Calabrese et al. 2011	Italy	Cohort	213	168	MS= 32.3 ± 7.2 HC=34.4 ± 7.9	MS=120 HC=32	6.8 ± 2.4	RRMS	1.5(0–4)	6
Calabrese et al. 2012	Italy	Cohort	312	312	RRMS=37.4 ± 7.3 PPMS=41.1 ± 5.0 SPMS=41.1 ± 4.1	196	4.1 ± 3.0	RRMS-PPMS-SPMS	2.0(1.0–5.5) and 4.0 (3.0–6.5)	5
Filippi et al. 1997	Italy	Cohort	14	14	RRMS=29±5.8 SPMS=41±9.2	RRMS=4 SPMS=4	RRMS=6 SPMS=10	RRMS-SPMS	RRMS= 1.5(1.0–3.5), SPMS=6.0(5.0–6.0)	8
Hasan et al. 2012	USA	Cohort	136	68	MS=41.6 ± 8.5 HC=40.0 ± 8.6	MS=53 HC=39	8.4	RRMS	1.5(0–4.5)	7
Horakova et al. 2009	Czech Republic	Cohort	208	181	30.7 ± 7.9	142	5.5 ± 5.2	NR	1.9 ± 0.9	6
Schreiber et al. 2001	Denmark	Cohort	86	86	43.3 ± 4.3	NR	13.6	RRMS-PPMS-SPMS	6.0 (1–9)	7
Tauhid et al. 2015	USA	Cohort	100	100	45.5 ± 9.7	75	12.1 ± 8.1 (1.6–36.6) years	CIS-RRMS-SPMS-PPMS	2.2 ± 1.9	8
Tavazzi et al. 2007	Italy	Cohort	470	432	44.4 ± 10.2	346	12±8.6	RRMS-SPMS-PPMS-CIS	3.3 ± 2	7
Sowa et al. 2019	Norway	Cohort	77	77	39.9 ± 10.3	60	9	RRMS	2.0 (1–2.75)	7
Dwyer et al. 2019	USA	Cohort	655	579	43.7 ± 8.1	507	11.2 ± 6.8	RRMS-SPMS-PPMS-CIS	2.5 (2.5)	7
Dwyer et al. 2021	USA	Cohort	1849	1849	MS=41.2 ± 13.8 HC=39.5 ± 9.4	1341	12.3 ± 13.2 months	RRMS-SPMS-PPMS	2.5(1.0–6.0)	7
Al-Radaideh et al. 2021	Jordan	Cross Sectional	60	30	RRMS= 31.27 ± 10.2 HC=32.35 ± 8.7	RRMS= 17 HC=16	46.03±14.10	RRMS	2.67 (0–6.5)	7
Bakshi et al. 2014	USA	Cross Sectional	85	55	MS= 41.1 ± 9.0 control=43.9 ± 6.3	MS=38 control=21	8.3 ± 7.4	RRMS-SPMS-PPMS-CIS	1.6 ± 1.7	7
Baranovicova et al. 2016	Slovakia	Cross Sectional	56	29	MS= 36.9 HC= 38.11	MS= 19 HC= 17	8.25	NR	3.44(1–8)	8
Barreiro-Gonzalez et al. 2022	Spain	Cross Sectional	64	64	45.1 ± 10.6	49	15.5 ± 8.2	RRMS-SPMS	3(1–7)	6
Bergsland et al. 2012	USA	Cross Sectional	401	401	CIS=29.3 RRMS=30.7		47 months	RRMS-CIS	2	8
Bergsland et al. 2021	USA	Cross Sectional	156	108	MS= 51.2 ± 10.4 HC=50±14.8	MS=78 HC=35	19.3 ± 10.0	RRMS-SPMS-PPMS	3.0 (1.5–5.9)	7
Calabrese et al. 2007	Italy	Cross Sectional	155	115	MS= 36.2 ± 11.3 HC= 36.2 ± 10.2	MS=66 HC=25	4.8 ± 5.2	RRMS-SPMS-CIS	2.3(0–7.0)	6

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Table 1 (Continued)

Author, year	Country	Type of study	Sample size	Number of MS	mean age of MS	Number of Female	Disease duration	Type of MS	EDSS mean baseline	NOS
Calabrese et al. 2007	Italy	Cross Sectional	420	380	36.2 ± 7.2	MS=284 HC=30	5.6 ± 5.1	RRMS-SPMS-CIS	2.7(0–7.0)	6
Ceccarelli et al. 2009	USA	Cross Sectional	86	60	SPMS=43.6 ± 9 HC=40.7 ± 9.8	51	23.11±5 months	SPMS	1.5(0–3)	7
Choi et al. 2020	USA	Cross Sectional	49	49	45.4 ± 11.2	51	NR	RRMS-SPMS-PPMS	5(1–6.5)	7
Choi et al. 2021	USA	Cross Sectional	47	47	46.2 ± 11.2	30	11.5 ± 8.3	RRMS-PMS	3.0(1–6.5)	6
Chou et al. 2018	UK	Cross Sectional	31	21	36 ± 10	17	65 months	CIS-MS	3.0 (3.0–3.5)	8
Chu et al. 2016	USA	Cross Sectional	35	26	MS=18 HC=7	MS=43 HC=45	9.56±8.60	NR	1.98 (0–6.5)	7
Chu et al. 2017	USA	Cross Sectional	29	14	MS=50.2 ± 8.2 HC=37.7 ± 9.6	MS=11 HC=9	18.4 ± 10.7	SPMS-PPMS-CIS	3.1(0–6)	7
Cohen et al. 2012	USA	Cross Sectional	21	21	40.9 ± 8	NR	8.3 ± 7.5	RRMS-SPMS-PPMS-CIS	1.6(0–6)	8
Dastidar et al. 1999	Finland	Cross Sectional	28	28	46	14	NR	SPMS	4.8(3.5–6.5)	6
Datta et al. 2015	USA	Cross Sectional	1424	924	MS=37.53±9.65 HC=37.68±10.93	674	583±1168 days	RRMS	1.94(0–6)	7
Dell'Oglio et al. 2015	USA	Cross Sectional	91	61	MS=41.0 ± 8.6 HC=43.9 ± 6.3	63	8.3 ± 7.2	RRMS-SPMS-PPMS-CIS	1.6(0–8.0)	6
Edwards et al. 2001	UK	Cross Sectional	40	40	RRMS=32.90 SPMS=40.10	27	5.6	RRMS-SPMS	1.68(0–3.5)	8
Fenu et al. 2018	Italy	Cross Sectional	80	60	MS=38.66±10.91 HC=38.02±9.97	50	12.9 ± 9.9	NR	2.33±1.57	6
Galego et al. 2015	Portugal	Cross Sectional	19	19	55.7 ± 7.6	14	13.0 ± 8.8	NR	6(3.5–8)	6
Ge et al. 2001	USA	Cross Sectional	36	18	MS=34.1 HC=33.9	MS=15 HC=12	5.3	NR	(0–6.5)	7
Giugni et al. 1997	Italy	Cross Sectional	85	85	RRMS=29.7 ± 7.5 SPMS=39.6 ± 7.8	RRMS=36 SPMS=22	RRMS=4.9 ± 2.9 SPMS=10.3 ± 5.6	RRMS-SPMS	1.9(0–3.5)	7
Gonyea et al. 2015	USA	Cross Sectional	47	23	MS=43±9 HC=44±11	MS=17 HC=11	6.1	RRMS-CIS	1.7 ± 0.9	7
Gracien et al. 2016	Germany	Cross Sectional	32	22	MS=34.7 ± 8.72 HC=34.2 ± 10.79	MS=2 HC=8	4 ± 6.54	RRMS	1.4(0–3)	6
Gracien et al. 2016	Germany	Cross Sectional	22	11	MS=46.8 ± 11.03 HC=43.6 ± 11.17	MS=7 HC=6	15.3 ± 6.75	SPMS	5.8(3–8.5)	7
Grimaud et al. 1999	UK	Cross Sectional	15	15	36	NR	NR	MS-RRMS-SPMS	4.5(1.0–9.5)	7
Hannoun et al. 2012	France	Cross Sectional	68	41	41.7 ± 5.2	RRMS=19 SPMS=7 HC=15	RRMS=5.5 ± 2.6 SPMS=15.9 ± 4.3	RRMS-SPMS	RRMS=2.4(0.0–4.5), SPMS=5.4(4.0–7.0)	7
Hemond et al. 2018	USA	Cross Sectional	91	61	43.9 ± 6.3	MS=42 HC=21	8.3 ± 7.2	RRMS-SPMS-PPMS-CIS	1.6 ± 1.7	7
Hickman et al. 2001	UK	Cross Sectional	9	9	42	NR	13	RRMS-SPMS	8(3–8.5)	8
Kearney et al. 2014	UK	Cross Sectional	159	159	52±8.8	111	26.2 ± 6.7	RRMS-SPMS	4(0–8)	8
Kocsis et al. 2021	Italy	Cross Sectional	79	79	42.27±9.89	55	12.3 ± 7.4	RRMS	2(0–6)	5
Kolber et al. 2015	Germany	Cross Sectional	122	122	31.75	85	1.65	RRMS-CIS	1.0(0.0–4.0)	7
Nijeholt et al. 1998	Netherlands	Cross Sectional	91	91	35	54	5	RRMS-SPMS-PPMS	1 (0–4.5)	6
Ramanathan et al. 2013	Australia	Cross Sectional	63	63	40	44	3	RRMS-SPMS	1.5 (1.0–3.5)	8
Sastre-Garriga et al. 2005	UK	Cross Sectional	31	31	46	13	3	PPMS	4.5 (3.5–7)	8
Senda et al. 2012	Japan	Cross Sectional	50	32	46.5 ± 8.8	25	10.1	RRMS-PPMS	3.5 (0–8.5)	8
Singhal et al. 2019	USA	Cross Sectional	17	12	42±11.7	7	12.7	RRMS-SPMS	3.5(1–6.5)	7
Traboulsee et al. 2003	Canada	Cross Sectional	158	95	41.5 ± 8.3	27	10.7 ± 7.7	RRMS-SPMS	2.0 ± 2.5	7
Ukkonen et al. 2003	Finland	Cross Sectional	48	28	51±9	14	12 ± 9	PPMS	4.9 (2.0–8.0)	4
Vidal-Jordana et al. 2018	Canada	Cross Sectional	90	90	38.0 ± 10.2	61	8.1 ± 8.8		2 ± 1.5	8

(continued on next page)

Table 1 (Continued)

Author, year	Country	Type of study	Sample size	Number of MS	mean age of MS	Number of Female	Disease duration	Type of MS	EDSS mean baseline	NOS
Caramanos et al. 2012	Canada	Cross-Sectional, Retrospective Study	110	110	RRMS=37.93±8.74 SPMS=46.78±11.4	77	RRMS=8.99±7.42 SPMS=16.80±7.97	RRMS-SPMS	2.44 (0–7.5)	6
Calabrese et al. 2009	Italy	Longitudinal Study	70	48	MS=41.2 ± 5.1 HC=35.1 ± 7.1	MS=28 HC=15	4.0 ± 3.2	PPMS	4.0 (3.0–6.5)	7
Calabrese et al. 2012	Italy	Longitudinal Study	120	92	MS=37.4 ± 7.2 HC=15.4 ± 3.8	MS=13.3 ± 2.8 HC=221	1.2	NR	3.2 (1.0–7.0)	6
De Meo et al. 2019	Italy	Longitudinal Study	411	68	MS=15.1 ± 2.2 HC=33.2 ± 7.2	221	0.32±0.11 4 ± 5 3	NR	1.5 (1.0–1.5)	4
Cappelle et al. 2020	Spain	Cross Sectional	132	132	40.7 ± 9.1	79	8.3 ± 7.0	RRMS-SPMS-PPMS-CIS	1.5 (1–5)	8
Yousuf et al. 2016	USA	Cross Sectional	71	51		35			1.6 ± 1.7	6

NR, Not Reported; EDSS, Expanded Disability Status Scale; RRMS, relapsing remitting multiple sclerosis; SPMS, secondary progressive multiple sclerosis; PPMS, primary progressive multiple sclerosis; PRMS, progressive relapsing multiple sclerosis; CIS, Clinically Isolated Syndrome; NOS, Newcastle-Ottawa Scale.

df = 21, I^2 = 80 %) (Fig. 4). We were unable to perform the same analysis on cortical T2-Lesion volume due to the low number of studies.

Five studies were included in the analysis of WM lesion volume. No significant correlations were found (95 % CI: −0.16 to 0.56; z-score: 0.23, χ^2 = 30.14, df = 4, I^2 = 87 %).

Disability and regional brain volumes

Investigating the correlation between brain parenchymal fraction and EDSS score yielded a significant result (95 % CI: −0.35 to −0.24; z-score: −0.30) along with moderate (χ^2 = 37.81, df = 19, I^2 = 50). Moreover, cortical thickness was negatively associated with the level of disability in patients with MS (95 % CI: −0.38 to −0.12; z-score: −0.25, χ^2 = 0.47, df = 1, I^2 = 0 %).

Also, we investigated the correlation between disability and frontal atrophy which resulted in a significant positive correlation (95 % CI: 0.14 to 0.39; z-score: 0.27, χ^2 = 0.03, df = 1, I^2 = 0 %). Furthermore, two studies reported the correlation between changes in the frontal gyrus and EDSS score. Pooled analysis revealed a significant negative correlation (95 % CI: −0.60 to −0.37; z-score: −0.50, χ^2 = 0.25, df = 1, I^2 = 0 %). However, there was no correlation between precentral gyrus volume (95 % CI: −0.69 to 0.24; z-score: −0.29, χ^2 = 4.35, df = 1, I^2 = 77 %).

We examined the correlation between the amygdala and EDSS score by including four studies. This analysis resulted in no significant correlations (95 % CI: −0.16 to 0.06; z-score: −0.05) along with moderate heterogeneity (χ^2 = 6.01, df = 3, I^2 = 50 %).

Pooling the reported correlations between hippocampus volume and disability assessed by EDSS among MS patients indicate that there was no significant correlation (95 % CI: −0.08 to 0.15; z-score: 0.04, χ^2 = 0.32, df = 2, I^2 = 0 %).

We investigated the potential association between thalamus volume and EDSS score. We found a significant negative correlation (95 % CI: −0.34 to −0.10; z-score: −0.22, χ^2 = 136.23, df = 17, I^2 = 88 %).

We pooled the results of six studies to assess the correlation of corpus callosum volume and disability in patients with MS. There was no significant association (95 % CI: −0.47 to 0.15; z-score: −0.17, χ^2 = 29.78, df = 5, I^2 = 83 %). Similarly, there was a non-significant association between the volume of globus pallidus and EDSS score based on our meta-analysis (95 % CI: −0.29 to 0.07; z-score: −0.11, χ^2 = 49.89, df = 8, I^2 = 84 %). Furthermore, the volume of nucleus accumbens was not correlated with the EDSS score (95 % CI: −0.07 to 0.17; z-score: 0.05). Moreover, there was no heterogeneity among studies (χ^2 = 0.22, df = 1, I^2 = 0 %).

Furthermore, we investigated the association between EDSS scores in patients with MS and the volume of putamen by pooling the results of nine studies. The results showed a significant correlation (95 % CI: −0.41 to −0.09; z-score: −0.26) and a substantial heterogeneity among studies (χ^2 = 34.62, df = 8, I^2 = 77 %).

Eight of the included studies were analyzed to measure the correlation between caudate nucleus volume and disability. The meta-analysis showed a significant correlation between caudate nucleus volume and EDSS score (95 % CI: −0.31 to −0.03; z-score: −0.17, χ^2 = 14.43, df = 7, I^2 = 52 %). Moreover, a positive correlation between infratentorial characteristics and EDSS was observed (95 % CI: 0.13 to 0.70; z-score: 0.46), χ^2 = 13.82, df = 5, I^2 = 64 %).

We investigated the characteristics of ventricles in the brain and their possible association with disability in MS patients. Our analysis showed that there was no significant correlation between lateral ventricular volume and EDSS score (95 % CI: −0.03 to 0.04; z-score: 0.0, χ^2 = 0.01, df = 1, I^2 = 0 %). Analysis based on third ventricle width by including three studies showed a significant correlation (95 % CI: 0.08 to 0.56; z-score: 0.34, χ^2 = 5.50, df = 2, I^2 = 64 %).



Fig. 2. Significant (S, red), and non-significant (N, green) correlations between EDSS score and MRI features.

Disability and cerebellum volume

In addition, the potential association between cerebellum volume and EDSS score in MS patients was investigated. Meta-analysis revealed no significant correlations between cerebellum volume and level of disability among these patients (95 % CI: -0.71 to 0.39 ; z-score: -0.24). The heterogeneity of studies was high ($\chi^2 = 52.90$, $df = 5$, $I^2 = 91\%$). Similar to the cerebellum, the volume of pons was not correlated with EDSS among patients with MS (95 % CI: -0.48 to 0.09 ; z-score: -0.21 , $\chi^2 = 2.84$, $df = 1$, $I^2 = 65\%$).

Discussion

The EDSS is a widely used tool for measuring disease outcomes in clinical trials and clinical practice for patients with MS.¹³⁸ It is based on a standardized neurological examination and assesses seven functional systems: visual, brainstem, pyramidal, cerebellar, sensory, bowel-bladder, and cerebral functions. The EDSS is used to assess

disability and monitor the effectiveness of therapies.¹³⁹ Previous systematic reviews and meta-analyses have investigated the correlations between certain MRI indices and the EDSS in MS patients.^{28,140,141} For example, Altokhis et al. conducted a systematic review and meta-analysis of 15 studies, finding a significant association between the number and volume of white matter lesions and disability progression after 10 years based on EDSS scores.¹⁴⁰ The study revealed that having four or more lesions detected in the baseline MRI was significantly associated with EDSS 3, with an odds ratio of 4.1, and with EDSS 6, with an odds ratio of 4.3. Similarly, having ten or more lesions was associated with odds ratios of 4.15 for EDSS 3 and 5.54 for EDSS 6.¹⁴⁰ Furthermore, Valizadeh et al. conducted a meta-analysis including 27 studies and a total of 1919 patients, which found a significant correlation between the mean volume of T1 hypointense lesions and EDSS score, with a correlation coefficient of 0.32.²⁸ To address the limitations of previous meta-analyses, such as the limited number of included studies and analyzed MRI parameters, we conducted a comprehensive systematic review that included

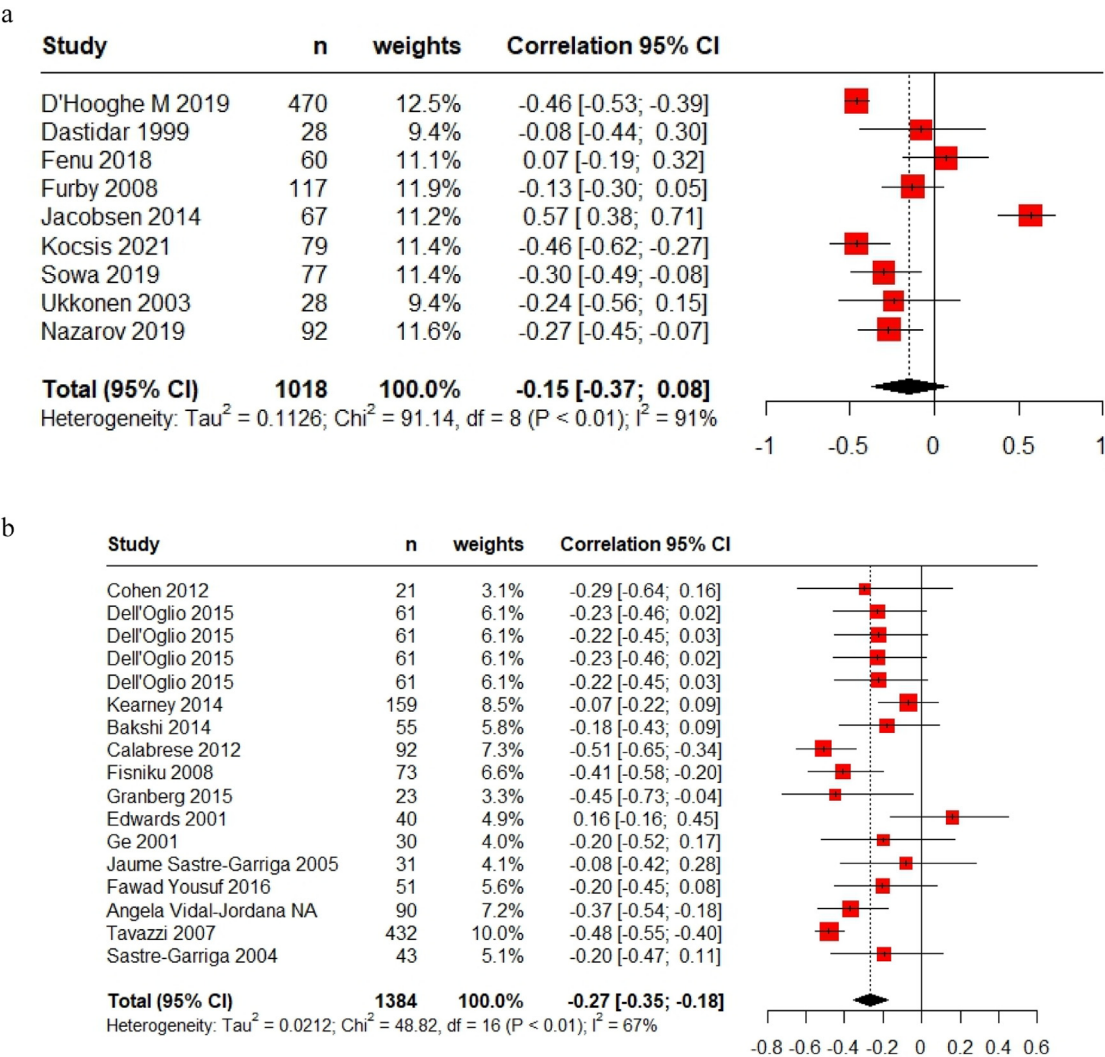


Fig. 3. Correlation of EDSS score and MRI features. (a) Brain volume, (b) GM fraction.

studies examining various parameters of T1-weighted and T2-weighted scans.

In the current meta-analysis, significant negative correlations were observed between the EDSS score and various factors, including white matter volume, gray matter volume, caudate nucleus volume, brain parenchymal fraction, gray matter fraction, white matter fraction, normal-appearing white matter, cortical thickness, and changes in the thalamus and putamen. Additionally, the EDSS score showed positive correlations with total lesion volume, cortical lesion volume, T1-Lesion volume, and T2-Lesion volume, as well as cortical T1, frontal atrophy, infratentorial characteristics, and third ventricle volumes.

Gray matter atrophy, which indicates axonal loss and irreversible neuronal damage, is a common characteristic of MS and has been linked to cognitive decline and disability.¹⁴² Among the different gray matter volumes, our findings revealed that atrophy in the thalamus, putamen, caudate nucleus, cortex, and frontal regions was associated with higher EDSS scores. Thalamic atrophy is thought to result from widespread inflammatory processes and demyelination. The thalamus, due to its extensive connections with various brain regions, may be particularly susceptible to axonal damage, leading to more pronounced atrophy.²³ In contrast, some studies have reported no significant association between thalamic changes and the EDSS score. This discrepancy could be attributed to variations in the techniques used to measure volumes of deep gray matter

(e.g., manual, semi-automated, or fully automated methods), differences in the contrast-to-noise ratio of the MRI data, and variations in image alterations caused by demyelination and/or localized iron deposition.^{143,144}

In addition, the caudate nucleus is involved in various functions, including working memory, cognitive function, emotions, learning, memory, reward, motivation, and romantic interaction.¹⁴⁵ Previous studies have reported that caudate nucleus lesions in MS patients are associated with abulia, memory dysfunction, and fatigue.^{146,147} These findings align with our results, which suggest that lower caudate nucleus volumes are associated with higher disability. Furthermore, the putamen plays a role in diverse aspects such as learning, motor control, speech articulation, language abilities, reward processing, cognitive function, and addiction.¹⁴⁸ Putamen degeneration has been linked to deficits in executive functions and working memory in MS patients.¹⁴⁹ Another study involving 68 patients with relapsing-remitting MS and 26 healthy controls found that putamen degeneration begins shortly after the initial appearance of symptoms, or even years prior, and continues to worsen over time.¹⁵⁰

The infratentorial area of the brain refers to the region situated beneath the tentorium cerebelli, encompassing the cerebellum, brain stem, and the fourth ventricle.¹⁵¹ Although less common than supratentorial lesions, infratentorial lesions in patients with MS can still significantly impact clinical disability and cognitive impairment.

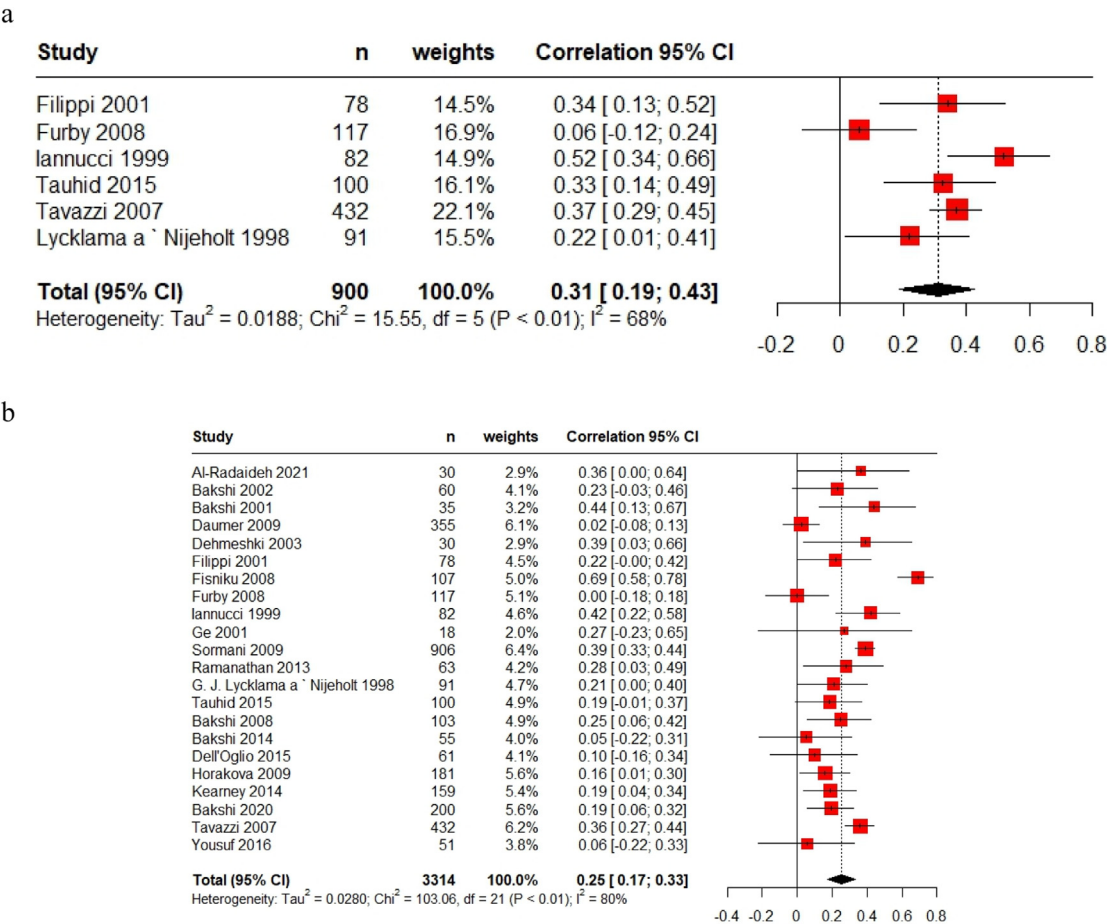


Fig. 4. Correlation of EDSS score and MRI features. (a) T1-Lesion, (b) T2-Lesion volume.

These lesions are considered to be relatively specific to MS, and their presence can aid in predicting long-term disability in individuals displaying initial signs indicative of the disease.^{152,153} In a 30-year follow-up study involving 132 participants with the clinically isolated syndrome, 80 developed MS, and 26 progressed to secondary progressive MS. Notably, the presence of infratentorial lesions at baseline was the most robust early predictor of secondary progressive MS (OR = 26.0, 95 % CI = 3.1–215.0) and an Expanded Disability Status Scale (EDSS) score greater than 3.5 (OR = 16.8, 95 % CI = 2.0–139.7) at the 30-year mark.¹⁵⁴ In our meta-analysis, we found a positive correlation between infratentorial lesions and EDSS scores. The correlation coefficient calculated for these studies falls within a 95 % confidence interval of 0.13 to 0.70, with a corresponding z-score of 1.96. However, given the substantial heterogeneity observed, further research is necessary to validate these findings and obtain a more comprehensive understanding of the relationship between infratentorial characteristics and EDSS in MS.

MS is characterized by widespread damage to the central nervous system, including changes in both normal-appearing white matter and demyelinating white matter lesions.¹⁵⁵ T2-weighted MRI scans are commonly utilized for diagnosing and managing MS. These scans are sensitive to fluid presence, enabling the identification of MS lesions in the brain and spinal cord.¹⁵⁶ Extensive research has been conducted on the number and characteristics of T2 lesions in MS patients, as well as their relationship with disease progression and disability.^{157–159} Our analysis, which incorporated 22 studies, demonstrated a significant correlation between T2-Lesion volume and EDSS scores. These findings indicate that a higher burden of T2 hyperintense lesions may be associated with increased disability in

MS patients. However, it is essential to consider the limitations of T2-weighted MRI in studies involving MS patients. Firstly, T2-weighted MRI images may display areas of hyperintensity, suggesting lesion presence. However, these hyperintense areas can also arise from other conditions or be unrelated to MS, such as normal aging,¹⁶⁰ hypertension,¹⁶¹ and diabetes.¹⁶² This lack of specificity can lead to misinterpretation or misdiagnosis.¹⁶³ Secondly, some studies have identified a "clinic-radiological paradox," referring to the weak correlation between the number and volume of T2 lesions and the progression of disability in MS patients over the medium to long term.¹⁶⁴ This could be due to the fact that T2-Lesion volumes represent various processes, such as edema, inflammation, formation of new or growing lesions, remyelination, or tissue degeneration.^{159,165} Thirdly, while T2-weighted images are sensitive to detecting MS lesions, they may not capture all lesions, particularly smaller or more subtle ones. Lesions situated in specific areas, such as the posterior fossa or spinal cord, can be particularly challenging to visualize using T2-weighted images alone. Additional imaging sequences and techniques may be necessary to improve lesion detection and visualization in such cases.¹⁶⁶

T1-weighted MRI sequences are utilized to detect hypointense lesions in MS, commonly referred to as "black holes." These lesions signify areas of axonal loss and demyelination, potentially caused by localized hypoperfusion, and are believed to be irreversible.¹⁶⁷ Previous studies have proposed that T1 lesions may serve as better predictors of neurological disability in MS patients; however, the results remain controversial.^{168,169} In our current study, the analysis of T1-Lesion volume, involving six studies, revealed a significant positive correlation with EDSS scores. Nonetheless, substantial heterogeneity was observed

among these studies, indicating possible variations in imaging protocols or patient characteristics. Our findings align with a prior systematic review and meta-analysis that demonstrated a weak to moderate correlation, with a correlation coefficient of 0.32 (95 % CI 0.26–0.37), between the mean volume of T1 lesions and EDSS scores.²⁸ These findings suggest that relying solely on T1-weighted or T2-weighted lesion characteristics has limited efficacy in assessing disability in MS patients.

The T1/T2-weighted ratio has been proposed as a sensitive method for detecting subtle microstructural changes in the white matter of patients with MS.¹⁷⁰ A recent study involving 171 patients with clinically isolated syndrome and 22 MS patients found that T1-w/T2-w values decreased during the lesion development process, starting from normal-appearing white matter. Furthermore, the study revealed a negative correlation between T1-w/T2-w values and disability as well as disease duration.¹⁷¹ Moreover, several recent studies have developed machine learning and deep learning algorithms that exhibit higher accuracy in early lesion detection, extraction of lesion characteristics, and prediction of disease progression. These advancements hold the potential to provide insights into the mechanisms of MS and improve outcomes for patients.^{172–174}

Limitations

In the present study, we conducted a comprehensive systematic review of the literature and performed a meta-analysis incorporating various factors. However, there are several limitations that need to be considered. First, we observed considerable heterogeneity in many of the analyzed parameters, which can be attributed to several factors. For instance, variations in patient characteristics, such as disease subtype, duration, age, and sex, can contribute to differences in lesion patterns. Moreover, disease stage, activity, and treatment status can also impact lesion development and progression. Furthermore, differences in MRI protocols, lesion segmentation methods, and study designs further contribute to the observed heterogeneity. Additionally, we should acknowledge the potential for publication bias, as studies with positive findings may be more likely to be published. Second, while the EDSS is widely used, it has certain weaknesses. These weaknesses include limited inter- and intra-rater reliability, as well as limited sensitivity to change, especially in patients with higher initial scores.²⁷ Another limitation that should be stated is the inclusion of studies performed by brain imaging only and spinal cord imaging studies were excluded. Based on previous studies, spinal cord lesions can significantly contribute the disability in MS patients. A previous study by Nasiri et al. found that, late-onset MS patients have a high rate of spinal cord lesions and supratentorial involvement.¹⁷⁵ We observed a high level of heterogeneity in a number of analyses which might be due to the variation in clinical imaging and brain metrics, the effect of different treatments, and EDSS variability. However, lack of the appropriate information regarding the imaging techniques in some of the studies makes it impossible to perform subgroup analysis in order to decrease the level of heterogeneity.

Conclusion

In conclusion, this systematic review and meta-analysis revealed that disability in patients with MS is linked to extensive changes in different brain regions, encompassing gray and white matter, supratentorial and infratentorial structures, as well as T1 and T2 weighted MRI lesions. Nonetheless, it is crucial to acknowledge that there was heterogeneity observed among the studies included, highlighting the necessity for further research and validation of these findings. Overall, these findings enhance our understanding of the connection between MRI features and disability in MS, providing valuable insights that can assist in clinical decision-making and the development of targeted therapies for MS patients. Future studies investigating the possible association between MRI measures and other domains of disability such as cognitive impairment and upper limb

dysfunction can better illustrate the role of MRI in predicting specific types of disabilities in MS patients.

Compliance with ethics guidelines

This article is based on previously conducted studies and does not contain any new studies with human participants or animals performed by any of the authors.

Ethical approval

Not applicable.

Availability of data and material

The datasets analyzed during the current study are available upon request with no restriction.

Consent for publication

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Declaration of Competing Interest

The authors declare no conflict of interest regarding the publication of this paper.

CRediT authorship contribution statement

Fardin Nabizadeh: Conceptualization, Formal analysis, Writing – review & editing, Writing – original draft, Data curation, Methodology, Validation. **Rasa Zafari:** Writing – original draft, Data curation, Methodology, Validation. **Mobin Mohamadi:** Writing – review & editing. **Tahereh Maleki:** Conceptualization, Formal analysis, Writing – review & editing, Writing – original draft, Data curation, Methodology, Validation. **Mohammad Sadeq Fallahi:** Conceptualization, Formal analysis, Writing – review & editing. **Nazanin Rafiei:** Conceptualization, Formal analysis, Writing – review & editing, Writing – original draft, Data curation, Methodology, Validation.

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References

1. Kasper D, Fauci A, Hauser S, Longo D, Jameson J, Loscalzo J. *Harrison's Principles of Internal Medicine*. 19e. NY, USA: McGraw-hill New York; 2015.
2. Browne P, Chandraratna D, Angood C, Tremlett H, Baker C, Taylor BV, et al. Atlas of multiple sclerosis 2013: a growing global problem with widespread inequity. *Neurology*. 2014;83(11):1022–1024.
3. Global, regional, and national burden of multiple sclerosis 1990–2016: a systematic analysis for the global burden of disease study 2016. *Lancet Neurol*. 2019;18(3):269–85.
4. Hunter SF. Overview and diagnosis of multiple sclerosis. *Am J Manag Care*. 2016;22(6 Suppl):s141–s150.
5. Pathak L. Personalized treatment for multiple sclerosis: the role of precision medicine. *Neurol Lett*. 2023;2(1):30–34.
6. Hauser SL, Cree BAC. Treatment of multiple sclerosis: a review. *Am J Med*. 2020;133(12):1380–1390. e2.
7. Lublin FD, Reingold SC, Cohen JA, et al. Defining the clinical course of multiple sclerosis: the 2013 revisions. *Neurology*. 2014;83(3):278–286.
8. Bergamaschi R. Prognostic factors in multiple sclerosis. *Int Rev Neurobiol*. 2007;79:423–447.
9. Melcon MO, Correale J, Melcon CM. Is it time for a new global classification of multiple sclerosis? *J Neurol Sci*. 2014;344(1–2):171–181.

10. McGinley MP, Goldschmidt CH, Rae-Grant AD. Diagnosis and treatment of multiple sclerosis: a review. *JAMA*. 2021;325(8):765–779.
11. Mirmosayyeb O, Moases Ghaffary E. Optical Coherence Tomography (OCT) measurements and disability in Multiple Sclerosis (MS): protocol of systematic review. *Neurol Lett*. 2023;2(1):44–46.
12. Filippi M, Brück W, Chard D, et al. Association between pathological and MRI findings in multiple sclerosis. *Lancet Neurol*. 2019;18(2):198–210.
13. Thompson AJ, Banwell BL, Barkhof F, et al. Diagnosis of multiple sclerosis: 2017 revisions of the McDonald criteria. *Lancet Neurol*. 2018;17(2):162–173.
14. Rovaris M, Iannucci G, Falautano M, et al. Cognitive dysfunction in patients with mildly disabling relapsing-remitting multiple sclerosis: an exploratory study with diffusion tensor MR imaging. *J Neurol Sci*. 2002;195(2):103–109.
15. Vrenken H, Geurts JJ, Knol DL, et al. Whole-brain T1 mapping in multiple sclerosis: global changes of normal-appearing gray and white matter. *Radiology*. 2006;240(3):811–820.
16. Solomon AJ, Bourdette DN, Cross AH, et al. The contemporary spectrum of multiple sclerosis misdiagnosis: a multicenter study. *Neurology*. 2016;87(13):1393–1399.
17. Pirahesh K, Motamedi-Jahromi A. Acute relapse of Multiple Sclerosis (MS) in an adolescent patient after Tuberculin skin test (TST): a case report. *Neurol Lett*. 2022;1(1):34–36.
18. Azevedo CJ, Overton E, Khadka S, et al. Early CNS neurodegeneration in radiologically isolated syndrome. *Neurol Neuroimmunol Neuroinflamm*. 2015;2(3):e102.
19. Chataway J, De Angelis F, Connick P, et al. Efficacy of three neuroprotective drugs in secondary progressive multiple sclerosis (MS-SMART): a phase 2b, multiarm, double-blind, randomised placebo-controlled trial. *Lancet Neurol*. 2020;19(3):214–225.
20. Chataway J, Schuerer N, Alsanousi A, et al. Effect of high-dose simvastatin on brain atrophy and disability in secondary progressive multiple sclerosis (MS-STAT): a randomised, placebo-controlled, phase 2 trial. *Lancet*. 2014;383(9936):2213–2221.
21. Hwang K, Bertolero MA, Liu WB, D'Esposito M. The human thalamus is an integrative hub for functional brain networks. *J Neurosci*. 2017;37(23):5594–5607.
22. Lanciego JL, Luquin N, Obeso JA. Functional neuroanatomy of the basal ganglia. *Cold Spring Harb Perspect Med*. 2012;2(12):a009621.
23. Magon S, Tsagkas C, Gaetano L, et al. Volume loss in the deep gray matter and thalamic subnuclei: a longitudinal study on disability progression in multiple sclerosis. *J Neurol*. 2020;267(5):1536–1546.
24. Filippi M, Agosta F. Diffusion tensor imaging and functional MRI. *Handb Clin Neurol*. 2016;136:1065–1087.
25. Eshaghi A, Young AL, Wijeratne PA, et al. Identifying multiple sclerosis subtypes using unsupervised machine learning and MRI data. *Nat Commun*. 2021;12(1):2078.
26. Kurtzke JF. Rating neurologic impairment in multiple sclerosis: an expanded disability status scale (EDSS). *Neurology*. 1983;33(11):1444–1452.
27. Meyer-Moock S, Feng YS, Maeurer M, Dippel FW, Kohlmann T. Systematic literature review and validity evaluation of the Expanded Disability Status Scale (EDSS) and the Multiple Sclerosis Functional Composite (MSFC) in patients with multiple sclerosis. *BMC Neurol*. 2014;14:58.
28. Valizadeh A, Moassemi M, Barati E, Ali Sahraian M, Aghajani F, Fattahi MR. Correlation between the clinical disability and T1 hypointense lesions' volume in cerebral magnetic resonance imaging of multiple sclerosis patients: a systematic review and meta-analysis. *CNS Neurosci Ther*. 2021;27(11):1268–1280.
29. Al-Radaideh A, Athamneh I, Alabadi H, Hbabbih M. Deep gray matter changes in relapsing-remitting multiple sclerosis detected by multi-parametric, high-resolution magnetic resonance imaging (MRI). *Eur Radiol*. 2021;31(2):706–715.
30. Elliott C, Belachew S, Wolinsky JS, et al. Chronic white matter lesion activity predicts clinical progression in primary progressive multiple sclerosis. *Brain*. 2019;142(9):2787–2799.
31. Tsagkas C, Magon S, Gaetano L, et al. Spinal cord volume loss: a marker of disease progression in multiple sclerosis. *Neurology*. 2018;91(4):e349–e358.
32. Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med*. 2009;6(7):e1000097.
33. Lo CKL, Mertz D, Loeb M. Newcastle-ottawa scale: comparing reviewers' to authors' assessments. *BMC Med Res Methodol*. 2014;14(1):45.
34. Audoin B, Davies GR, Finisku L, Chard DT, Thompson AJ, Miller DH. Localization of grey matter atrophy in early RRMS: a longitudinal study. *J Neurol*. 2006;253(11):1495–1501.
35. Audoin B, Ibarrola D, Malikova I, et al. Onset and underpinnings of white matter atrophy at the very early stage of multiple sclerosis – a two-year longitudinal MRI/MRSI study of corpus callosum. *Mult Scler*. 2007;13(1):41–51.
36. Bagnato F, Franco G, Ye F, et al. Selective inversion recovery quantitative magnetization transfer imaging: toward a 3 T clinical application in multiple sclerosis. *Mult Scler*. 2020;26(4):457–467.
37. Bakshi R, Benedict RH, Bermel RA, et al. T2 hypointensity in the deep gray matter of patients with multiple sclerosis: a quantitative magnetic resonance imaging study. *Arch Neurol*. 2002;59(1):62–68.
38. Bakshi R, Benedict RH, Bermel RA, Jacobs L. Regional brain atrophy is associated with physical disability in multiple sclerosis: semiquantitative magnetic resonance imaging and relationship to clinical findings. *J Neuroimaging*. 2001;11(2):129–136.
39. Bakshi R, Healy BC, Dupuy SL, et al. Brain MRI predicts worsening multiple sclerosis disability over 5 years in the SUMMIT study. *J Neuroimaging*. 2020;30(2):212–218.
40. Bakshi R, Neema M, Healy BC, et al. Predicting clinical progression in multiple sclerosis with the magnetic resonance disease severity scale. *Arch Neurol*. 2008;65(11):1449–1453.
41. Bakshi R, Neema M, Tauhid S, et al. An expanded composite scale of MRI-defined disease severity in multiple sclerosis: MRDSS2. *Neuroreport*. 2014;25(14):1156–1161.
42. Baranovicova E, Mlynarik V, Kantorova E, Hnilicova P, Dobrota D. Quantitative evaluation of cerebral white matter in patients with multiple sclerosis using multicomponent T2 mapping. *Neurol Res*. 2016;38(5):389–396.
43. Barnett M, Bergsland N, Weinstock-Guttman B, et al. Brain atrophy and lesion burden are associated with disability progression in a multiple sclerosis real-world dataset using only T2-FLAIR: the NeuroSTREAM MSBase study. *Neuroimage-Clin*. 2021;32.
44. Barreiro-Gonzalez A, Sanz MT, Carratala-Bosca S, et al. Design and validation of an expanded disability status scale model in multiple sclerosis. *Eur Neurol*. 2022;85(2):112–121.
45. Beer A, Biberacher V, Schmidt P, et al. Tissue damage within normal appearing white matter in early multiple sclerosis: assessment by the ratio of T1- and T2-weighted MR image intensity. *J Neurol*. 2016;263(8):1495–1502.
46. Bergsland N, Benedict RH, Dwyer MG, et al. Thalamic nuclei volumes and their relationships to neuroperformance in multiple sclerosis: a cross-sectional structural MRI study. *J Magn Reson Imaging*. 2021;53(3):731–739.
47. Bergsland N, Horakova D, Dwyer MG, et al. Subcortical and cortical gray matter atrophy in a large sample of patients with clinically isolated syndrome and early relapsing-remitting multiple sclerosis. *Am J Neuroradiol*. 2012;33(8):1573–1578.
48. Bonneville F, Moriarty DM, Li BS, Babb JS, Grossman RI, Gonen O. Whole-brain N-acetylaspartate concentration: correlation with T2-weighted lesion volume and expanded disability status scale score in cases of relapsing-remitting multiple sclerosis. *AJNR Am J Neuroradiol*. 2002;23(3):371–375.
49. Brex PA, Ciccarelli O, O'Riordan JI, Sailer M, Thompson AJ, Miller DH. A longitudinal study of abnormalities on MRI and disability from multiple sclerosis. *New Engl J Med*. 2002;346(3):158–164.
50. Calabrese M, Atzori M, Bernardi V, et al. Cortical atrophy is relevant in multiple sclerosis at clinical onset. *J Neurol*. 2007;254(9):1212–1220.
51. Calabrese M, Bernardi V, Atzori M, et al. Effect of disease-modifying drugs on cortical lesions and atrophy in relapsing-remitting multiple sclerosis. *Mult Scler J*. 2012;18(4):418–424.
52. Calabrese M, De Stefano N, Atzori M, et al. Detection of cortical inflammatory lesions by double inversion recovery magnetic resonance imaging in patients with multiple sclerosis. *Arch Neurol*. 2007;64(10):1416–1422.
53. Calabrese M, Federle L, Bernardi V, et al. The association of intrathecal immunoglobulin synthesis and cortical lesions predicts disease activity in clinically isolated syndrome and early relapsing-remitting multiple sclerosis. *Mult Scler J*. 2012;18(2):174–180.
54. Calabrese M, Filippi M, Rovaris M, et al. Evidence for relative cortical sparing in benign multiple sclerosis: a longitudinal magnetic resonance imaging study. *Mult Scler*. 2009;15(1):36–41.
55. Calabrese M, Rinaldi F, Mattisi I, et al. The predictive value of gray matter atrophy in clinically isolated syndromes. *Neurology*. 2011;77(3):257–263.
56. Calabrese M, Rinaldi F, Poretto V, Gallo P. The puzzle of multiple sclerosis: gray matter finds its place. *Expert Rev Neurother*. 2011;11(11):1565–1568.
57. Calabrese M, Rocca M, Atzori M, et al. Cortical lesions in relapse-onset multiple sclerosis and disability: a 3-year longitudinal MRI study. *Mult Scler J*. 2009;15(9):S108–S.
58. Cappelle S, Pareto D, Tintore M, et al. A validation study of manual atrophy measures in patients with multiple sclerosis. *Neuroradiology*. 2020;62(8):955–964.
59. Caramanos Z, Francis SJ, Narayanan S, Lapierre Y, Arnold DL. Large, nonplateauing relationship between clinical disability and cerebral white matter lesion load in patients with multiple sclerosis. *Arch Neurol*. 2012;69(1):89–95.
60. Ceccarelli A, Filippi M, Neema M, et al. T2 hypointensity in the deep gray matter of patients with benign multiple sclerosis. *Mult Scler*. 2009;15(6):678–686.
61. Choi S, Li X, Mizell R, Spini M, Hua J, Harrison DM. Distribution-based measures of quantitative magnetic susceptibility and R2* from 7-Tesla MRI are associated with disability in multiple sclerosis. *Mult Scler J*. 2020;26(1_Suppl):87–88.
62. Choi S, Spini M, Hua J, Harrison DM. Blood-brain barrier breakdown in non-enhancing multiple sclerosis lesions detected by 7-Tesla MP2RAGE Delta T-1 mapping. *PLoS ONE*. 2021;16(4).
63. Chou JJ, Lim SY, Tanasescu R, et al. Seven-tesla magnetization transfer imaging to detect multiple sclerosis white matter lesions. *J Neuroimaging*. 2018;28(2):183–190.
64. Chu RX, Hurwitz S, Tauhid S, Bakshi R. Automated segmentation of cerebral deep gray matter from MRI scans: effect of field strength on sensitivity and reliability. *BMC Neurol*. 2017;17.
65. Chu RX, Tauhid S, Glanz BI, et al. Whole brain volume measured from 1.5T versus 3T MRI in healthy subjects and patients with multiple sclerosis. *J Neuroimaging*. 2016;26(1):62–67.
66. Coccozza S, Cosottini M, Signori A, et al. A clinically feasible 7-Tesla protocol for the identification of cortical lesions in multiple sclerosis. *Eur Radiol*. 2020;30(8):4586–4594.
67. Cohen AB, Neema M, Arora A, et al. The relationships among MRI-defined spinal cord involvement, brain involvement, and disability in multiple sclerosis. *J Neuroimaging*. 2012;22(2):122–128.
68. D'Hooghe MB, Gielen J, Van Remoortel A, et al. Single MRI-based volumetric assessment in clinical practice is associated with MS-related disability. *J Magn Reson Imaging*. 2019;49(5):1312–1321.

69. Dastidar P, Heinonen T, Lehtimäki T, et al. Volumes of brain atrophy and plaques correlated with neurological disability in secondary progressive multiple sclerosis. *J Neurol Sci*. 1999;165(1):36–42.
70. Datta S, Staewen TD, Cofield SS, et al. Regional gray matter atrophy in relapsing remitting multiple sclerosis: baseline analysis of multi-center data. *Mult Scler Relat Disord*. 2015;4(2):124–136.
71. Daumer M, Neuhaus A, Morrissey S, Hintzen R, Ebers GC. MRI as an outcome in multiple sclerosis clinical trials. *Neurology*. 2009;72(8):705–711.
72. De Meo E, Meani A, Moiola L, et al. Dynamic gray matter volume changes in pediatric multiple sclerosis: a 3.5 year MRI study. *Neurology*. 2019;92(15):e1709–e1723.
73. De Stefano N, Battaglini M, Smith SM. Measuring brain atrophy in multiple sclerosis. *J Neuroimaging*. 2007;17:105–155.
74. Dehmshki J, Chard DT, Leary SM, et al. The normal appearing grey matter in primary progressive multiple sclerosis - a magnetisation transfer imaging study. *J Neurol*. 2003;250(1):67–74.
75. Dell'Oglio E, Ceccarelli A, Glanz BI, et al. Quantification of global cerebral atrophy in multiple sclerosis from 3T MRI using SPM: the role of misclassification errors. *J Neuroimaging*. 2015;25(2):191–199.
76. DeStefano N, Fu L, Narayanan S, et al. Relationship between brain MRI and proton MR spectroscopy indices and clinical disability in a longitudinal study of patients with multiple sclerosis. *Neurology*. 1996;46(2):70006.
77. Dupuy SL, Tauhid S, Kim G, et al. MRI detection of hypointense brain lesions in patients with multiple sclerosis: T1 spin-echo vs. gradient-echo. *Eur J Radiol*. 2015;84(8):1564–1568.
78. Dwyer M, Lyman C, Ferrari H, et al. DeepGRAI (Deep gray rating via artificial intelligence): fast, feasible, and clinically relevant thalamic atrophy measurement on clinical quality T2-FLAIR MRI in multiple sclerosis. *NeuroImage Clin*. 2021;30.
79. Dwyer MG, Bergsland N, Ramasamy DP, Jakimovski D, Weinstock-Guttman B, Zivadinov R. Atrophied brain lesion volume: a new imaging biomarker in multiple sclerosis. *J Neuroimaging*. 2018;28(5):490–495.
80. Dwyer MG, Bergsland N, Ramasamy DP, et al. Salient central lesion volume: a standardized novel fully automated proxy for brain FLAIR lesion volume in multiple sclerosis. *J Neuroimaging*. 2019;29(5):615–623.
81. Edwards SGM, Liu C, Blumhardt LD. Cognitive correlates of supratentorial atrophy on MRI in multiple sclerosis. *Acta Neurol Scand*. 2001;104(4):214–223.
82. Elzayady M, Debees NL, Khalil M, Dawoud MM. Cerebellum and brain stem volume loss in relapsing remission multiple sclerosis by MRI volumetry: relation to neurological disability score and number of relapses. *Egypt J Radiol Nucl Med*. 2021;52(1).
83. Fenu G, Lorefice L, Loi L, et al. Adult brain volume in multiple sclerosis: the impact of paediatric onset. *Mult Scler Relat Disord*. 2018;21:103–107.
84. Filippi M. Linking structural, metabolic and functional changes in multiple sclerosis. *Eur J Neurol*. 2001;8(4):291–297.
85. Filippi M, Colombo B, Rovaris M, Pereira C, Martinelli V, Comi G. A longitudinal magnetic resonance imaging study of the cervical cord in multiple sclerosis. *J Neuroimaging*. 1997;7(2):78–80.
86. Fisher E, Rudick RA, Cutter G, et al. Relationship between brain atrophy and disability: an 8-year follow-up study of multiple sclerosis patients. *Mult Scler*. 2000;6(6):373–377.
87. Fisniku LK, Altmann DR, Cercignani M, et al. Magnetization transfer ratio abnormalities reflect clinically relevant grey matter damage in multiple sclerosis. *Mult Scler*. 2009;15(6):668–677.
88. Fisniku LK, Brex PA, Altmann DR, et al. Disability and T-2 MRI lesions: a 20-year follow-up of patients with relapse onset of multiple sclerosis. *Brain*. 2008;131:808–817.
89. Fisniku LK, Chard DT, Jackson JS, et al. Gray matter atrophy is related to long-term disability in multiple sclerosis. *Ann Neurol*. 2008;64(3):247–254.
90. Furby J, Hayton T, Anderson V, et al. Magnetic resonance imaging measures of brain and spinal cord atrophy correlate with clinical impairment in secondary progressive multiple sclerosis. *Mult Scler*. 2008;14(8):1068–1075.
91. Galego O, Gouveia A, Batista S, Moura C, Machado E. Brain atrophy and physical disability in primary progressive multiple sclerosis: a volumetric study. *Neuroradiol J*. 2015;28(3):354–358.
92. Ge Y, Grossman RI, Udupa JK, Babb JS, Kolson DL, McGowan JC. Magnetization transfer ratio histogram analysis of gray matter in relapsing-remitting multiple sclerosis. *Am J Neuroradiol*. 2001;22(3):470–475.
93. Ge Y, Grossman RI, Udupa JK, Babb JS, Nyúl LG, Kolson DL. Brain atrophy in relapsing-remitting multiple sclerosis: fractional volumetric analysis of gray matter and white matter. *Radiology*. 2001;220(3):606–610.
94. Ge Y, Grossman RI, Udupa JK, et al. Brain atrophy in relapsing-remitting multiple sclerosis and secondary progressive multiple sclerosis: longitudinal quantitative analysis. *Radiology*. 2000;214(3):665–670.
95. Giorgio A, Stromillo ML, Bartolozzi ML, et al. Relevance of hypointense brain MRI lesions for long-term worsening of clinical disability in relapsing multiple sclerosis. *Mult Scler*. 2014;20(2):214–219.
96. Giugni E, Pozzilli C, Bastianello S, et al. MRI measures and their relations with clinical disability in relapsing-remitting and secondary progressive multiple sclerosis. *Mult Scler*. 1997;3(4):221–225.
97. Gonyea JV, Watts R, Applebee A, et al. In vivo quantitative whole-brain T-1 rho MRI of multiple sclerosis. *J Magn Reson Imaging*. 2015;42(6):1623–1630.
98. Gracien RM, Jurcoane A, Wagner M, et al. Multimodal quantitative MRI assessment of cortical damage in relapsing-remitting multiple sclerosis. *J Magn Reson Imaging*. 2016;44(6):1600–1607.
99. Gracien RM, Jurcoane A, Wagner M, et al. The relationship between gray matter quantitative MRI and disability in secondary progressive multiple sclerosis. *PLoS ONE*. 2016;11(8).
100. Granberg T, Martola J, Bergendal G, et al. Corpus callosum atrophy is strongly associated with cognitive impairment in multiple sclerosis: results of a 17-year longitudinal study. *Mult Scler J*. 2015;21(9):1151–1158.
101. Grimaud J, Barker GJ, Wang L, et al. Correlation of magnetic resonance imaging parameters with clinical disability in multiple sclerosis: a preliminary study. *J Neurol*. 1999;246(10):961–967.
102. Hannoun S, Durand-Dubief F, Confavreux C, et al. Diffusion tensor-MRI evidence for extra-axonal neuronal degeneration in caudate and thalamic nuclei of patients with multiple sclerosis. *AJNR Am J Neuroradiol*. 2012;33(7):1363–1368.
103. Hardmeier M, Wagenpfeil S, Freitag P, et al. Atrophy is detectable within a 3-month period in untreated patients with active relapsing remitting multiple sclerosis. *Arch Neurol*. 2003;60(12):1736–1739.
104. Harrison DM, Roy S, Oh J, et al. Association of cortical lesion burden on 7-T magnetic resonance imaging with cognition and disability in multiple sclerosis. *JAMA Neurol*. 2015;72(9):1004–1012.
105. Hasan KM, Walimuni IS, Abid H, et al. Multimodal quantitative magnetic resonance imaging of thalamic development and aging across the human lifespan: implications to neurodegeneration in multiple sclerosis. *J Neurosci*. 2011;31(46):16826–16832.
106. Hasan KM, Walimuni IS, Abid H, Wolinsky JS, Narayana PA. Multi-modal quantitative MRI investigation of brain tissue neurodegeneration in multiple sclerosis. *J Magn Reson Imaging*. 2012;35(6):1300–1311.
107. Hemond CC, Chu R, Tummala S, Tauhid S, Healy BC, Bakshi R. Whole-brain atrophy assessed by proportional- versus registration-based pipelines from 3T MRI in multiple sclerosis. *Brain Behav*. 2018;8(8).
108. Hickman SJ, Brierley CM, Silver NC, et al. Infratentorial hypointense lesion volume on T1-weighted magnetic resonance imaging correlates with disability in patients with chronic cerebellar ataxia due to multiple sclerosis. *J Neurol Sci*. 2001;187(1–2):35–39.
109. Horakova D, Dwyer MG, Havrdova E, et al. Gray matter atrophy and disability progression in patients with early relapsing-remitting multiple sclerosis. A 5-year longitudinal study. *J Neurol Sci*. 2009;282(1–2):112–119.
110. Hurtado Rúa SM, Kaunzner UW, Pandya S, et al. Lesion features on magnetic resonance imaging discriminate multiple sclerosis patients. *Eur J Neurol*. 2022;29(1):237–246.
111. Iannucci G, Minicucci L, Rodegher M, Sormani MP, Comi G, Filippi M. Correlations between clinical and MRI involvement in multiple sclerosis: assessment using T-1, T-2 and MT histograms. *J Neurol Sci*. 1999;171(2):121–129.
112. Jacobsen C, Hagemeyer J, Myhr KM, et al. Brain atrophy and disability progression in multiple sclerosis patients: a 10-year follow-up study. *J Neurol Neurosurg Psychiatry*. 2014;85(10):1109–1115.
113. Kearney H, Rocca MA, Valsasina P, et al. Magnetic resonance imaging correlates of physical disability in relapse onset multiple sclerosis of long disease duration. *Mult Scler*. 2014;20(1):72–80.
114. Klistorner S, Barnett MH, Yiannikas C, et al. Expansion of chronic lesions is linked to disease progression in relapsing-remitting multiple sclerosis patients. *Mult Scler*. 2021;27(10):1533–1542.
115. Kocsis K, Szabó N, Tóth E, et al. Two classes of T1 hypointense lesions in multiple sclerosis with different clinical relevance. *Front Neurol*. 2021;12.
116. Kolber P, Montag S, Fleischer V, et al. Identification of cortical lesions using DIR and FLAIR in early stages of multiple sclerosis. *J Neurol*. 2015;262(6):1473–1482.
117. Lapucci C, Schiavi S, Signori A, et al. The role of disconnection in explaining disability in multiple sclerosis. *Eur Radiol Exp*. 2022;6(1).
118. Nazarov V, Makshakov G, Kalinin I, et al. Concentrations of immunoglobulin free light chains in cerebrospinal fluid predict increased level of brain atrophy in multiple sclerosis. *Immunol Res*. 2018;66(6):761–767.
119. Nijeholt G, van Walderveen MAA, Castelijns JA, et al. Brain and spinal cord abnormalities in multiple sclerosis - Correlation between MRI parameters, clinical subtypes and symptoms. *Brain*. 1998;121:687–697.
120. Pirpamer L, Kincses B, Kincses ZT, et al. Periventricular magnetisation transfer abnormalities in early multiple sclerosis. *NeuroImage-Clin*. 2022;34.
121. Ramanathan S, Lenton K, Burke T, et al. The utility of multimodal evoked potentials in multiple sclerosis prognostication. *J Clin Neurosci*. 2013;20(11):1576–1581.
122. Rizkallah M, Hefida M, Khalil M, Dawoud RM. Automated quantification of deep grey matter structures and white matter lesions using magnetic resonance imaging in relapsing remission multiple sclerosis. *Egypt J Radiol Nucl Med*. 2021;52(1).
123. Sastre-Garriga J, Ingle GT, Chard DT, et al. Grey and white matter volume changes in early primary progressive multiple sclerosis: a longitudinal study. *Brain*. 2005;128(6):1454–1460.
124. Sastre-Garriga J, Ingle GT, Ramio-Torrenta L, Miller DH, Thompson AJ. Grey and white matter atrophy in early clinical stages of primary progressive multiple sclerosis. *Neuroimage*. 2004;22(1):353–359.
125. Schreiber K, Sørensen PS, Koch-Henriksen N, et al. Correlations of brain MRI parameters to disability in multiple sclerosis. *Acta Neurol Scand*. 2001;104(1):24–30.

126. Senda J, Watanabe H, Tsuboi T, et al. MRI mean diffusivity detects widespread brain degeneration in multiple sclerosis. *J Neurol Sci.* 2012;319(1–2):105–110.
127. Singhal T, O'Connor K, Dubey S, et al. Gray matter microglial activation in relapsing vs progressive MS A F-18 PBR06-PET study. *Neurol-Neuroimmunol Neuroinflamm.* 2019;6(5).
128. Sormani MP, Rovaris M, Comi G, Filippi M. A reassessment of the plateauing relationship between T2 lesion load and disability in MS. *Neurology.* 2009;73(19):1538–1542.
129. Sowa P, Harbo HF, White NS, et al. Restriction spectrum imaging of white matter and its relation to neurological disability in multiple sclerosis. *Mult Scler J.* 2019;25(5):687–698.
130. Tauhid S, Chu R, Sasane R, et al. Brain MRI lesions and atrophy are associated with employment status in patients with multiple sclerosis. *J Neurol.* 2015;262(11):2425–2432.
131. Tavazzi E, Dwyer MG, Weinstock-Guttman B, et al. Quantitative diffusion weighted imaging measures in patients with multiple sclerosis. *Neuroimage.* 2007;36(3):746–754.
132. Traboulsee A, Dehmshki J, Peters KR, et al. Disability in multiple sclerosis is related to normal appearing brain tissue MTR histogram abnormalities. *Mult Scler J.* 2003;9(6):566–573.
133. Ukkonen M, Dastidar P, Heinonen T, Laasonen E, Elovaara I. Volumetric quantitation by MRI in primary progressive multiple sclerosis: volumes of plaques and atrophy correlated with neurological disability. *Eur J Neurol.* 2003;10(6):663–669.
134. Vaneckova M, Seidl Z, Krasensky J, et al. Patients' stratification and correlation of brain magnetic resonance imaging parameters with disability progression in multiple sclerosis. *Eur Neurol.* 2009;61(5):278–284.
135. Varoglu AO, Odaci E, Gumus H, Keles ON, Unal B, Deniz O. Evaluation of patients with multiple sclerosis using a combination of morphometrical features and clinical scores. *J Clin Neurosci.* 2010;17(2):191–195.
136. Vidal-Jordana A, Sastre-Garriga J, Pareto D, et al. Brain atrophy 15 years after CIS: baseline and follow-up clinico-radiological correlations. *Mult Scler J.* 2018;24(6):721–727.
137. Yousuf F, Kim G, Tauhid S, et al. The contribution of cortical lesions to a composite MRI scale of disease severity in multiple sclerosis. *Front Neurol.* 2016;7(JUN).
138. Treaba CA, Conti A, Klawiter EC, et al. Cortical and phase rim lesions on 7 T MRI as markers of multiple sclerosis disease progression. *Brain Commun.* 2021;3(3):fcab134.
139. Slezáková D, Kadlic P, Jezberová M, Boleková V, Valkovič P, Minar M. Brain volume loss in multiple sclerosis is independent of disease activity and might be prevented by early disease-modifying therapy. *Neurol Neurochir Pol.* 2023.
140. Altokhis AI, AlAmrani A, Alotaibi A, Podlasek A, Constantinescu CS. Magnetic resonance imaging as a prognostic disability marker in clinically isolated syndrome and multiple sclerosis: a systematic review and meta-analysis. *Diagnostics (Basel).* 2022;12(2).
141. Pike AR, James GA, Drew PD, Archer RL. Neuroimaging predictors of longitudinal disability and cognition outcomes in multiple sclerosis patients: a systematic review and meta-analysis. *Mult Scler Relat Disord.* 2022;57: 103452.
142. Lie IA, Weeda MM, Mattiesing RM, et al. Relationship between white matter lesions and gray matter atrophy in multiple sclerosis: a systematic review. *Neurology.* 2022;98(15):e1562–e1573.
143. Houtchens MK, Benedict RH, Killiany R, et al. Thalamic atrophy and cognition in multiple sclerosis. *Neurology.* 2007;69(12):1213–1223.
144. Prinster A, Quarantelli M, Orefice G, et al. Grey matter loss in relapsing-remitting multiple sclerosis: a voxel-based morphometry study. *Neuroimage.* 2006;29(3):859–867.
145. Driscoll ME, Bollu PC, Tadi P. *Neuroanatomy, Nucleus Caudate. StatPearls.* Treasure Island (FL): StatPearls Publishing; 2023.
146. Copyright © 2023, StatPearls Publishing LLC; 2023.
147. Hashiguchi S, Ogasawara N, Mine H, Kawachi Y. Multiple sclerosis with caudate lesions on MRI. *Intern Med.* 2001;40(4):358–362.
148. Ntranos A, Krieger S, Fabian M, et al. Fatigue in early multiple sclerosis and the role of the caudate nucleus. (P4.412). *Neurology.* 2018;90(15 Supplement). P4.412.
149. Ghandili M, Munakomi S. *Neuroanatomy, Putamen. StatPearls.* Treasure Island (FL): StatPearls Publishing; 2023.
150. Copyright © 2023, StatPearls publishing LLC; 2023.
151. Debernard L, Melzer TR, Alla S, et al. Deep grey matter MRI abnormalities and cognitive function in relapsing-remitting multiple sclerosis. *Psychiatry Res.* 2015;234(3):352–361.
152. Krämer J, Meuth SG, Tenberge JG, Schiffler P, Wiendl H, Deppe M. Early and degenerative putamen atrophy in multiple sclerosis. *Int J Mol Sci.* 2015;16(10):23195–23209.
153. Board PDQTE. *Childhood Central Nervous System Atypical Teratoid/Rhabdoid Tumor Treatment (PDQ®): Health Professional Version. PDQ Cancer Information Summaries.* Bethesda (MD): National Cancer Institute (US); 2002.
154. Minneboo A, Barkhof F, Polman CH, Uitdehaag BM, Knol DL, Castelijns JA. Infratentorial lesions predict long-term disability in patients with initial findings suggestive of multiple sclerosis. *Arch Neurol.* 2004;61(2):217–221.
155. Trip SA, Miller DH. Imaging in multiple sclerosis. *J Neurol Neurosurg Psychiatry.* 2005;76(Suppl 3):iii11–iii18. Suppl 3.
156. Chung KK, Altmann D, Barkhof F, et al. A 30-year clinical and magnetic resonance imaging observational study of multiple sclerosis and clinically isolated syndromes. *Ann Neurol.* 2020;87(1):63–74.
157. Alotaibi A, Podlasek A, Altokhis A, Aldhebaib A, Dineen RA, Constantinescu CS. Investigating microstructural changes in white matter in multiple sclerosis: a systematic review and meta-analysis of neurite orientation dispersion and density imaging. *Brain Sci.* 2021;11(9).
158. Preziosa P, Bouman PM, Kiljan S, et al. Neurite density explains cortical T1-weighted/T2-weighted ratio in multiple sclerosis. *J Neurol Neurosurg Psychiatry.* 2021;92(7):790–792.
159. Barnett M, Bergsland N, Weinstock-Guttman B, et al. Brain atrophy and lesion burden are associated with disability progression in a multiple sclerosis real-world dataset using only T2-FLAIR: the NeuroSTREAM MSBase study. *Neuroimage Clin.* 2021;32: 102802.
160. Eshaghi A, Marinescu RV, Young AL, et al. Progression of regional grey matter atrophy in multiple sclerosis. *Brain.* 2018;141(6):1665–1677.
161. Oship D, Jakimovski D, Bergsland N, et al. Assessment of T2 lesion-based disease activity volume outcomes in predicting disease progression in multiple sclerosis over 10 years. *Mult Scler Relat Disord.* 2022;67: 104187.
162. Haller S, Kövari E, Herrmann FR, et al. Do brain T2/FLAIR white matter hyperintensities correspond to myelin loss in normal aging? A radiologic-neuropathologic correlation study. *Acta Neuropathol Commun.* 2013;1:14.
163. Zhao Y, Ke Z, He W, Cai Z. Volume of white matter hyperintensities increases with blood pressure in patients with hypertension. *J Int Med Res.* 2019;47(8):3681–3689.
164. Wang DQ, Wang L, Wei MM, et al. Relationship between type 2 diabetes and white matter hyperintensity: a systematic review. *Front Endocrinol (Lausanne).* 2020;11: 595962.
165. Zhang Y, Zhu H, Mitchell JR, Costello F, Metz LM. T2 MRI texture analysis is a sensitive measure of tissue injury and recovery resulting from acute inflammatory lesions in multiple sclerosis. *Neuroimage.* 2009;47(1):107–111.
166. Louapre C. Conventional and advanced MRI in multiple sclerosis. *Rev Neurol (Paris).* 2018;174(6):391–397.
167. Thaler C, Faizy T, Sedlacik J, et al. T1- thresholds in black holes increase clinical-radiological correlation in multiple sclerosis patients. *PLoS ONE.* 2015;10(12): e0144693.
168. Erbayat Altay E, Fisher E, Jones SE, Hara-Cleaver C, Lee JC, Rudick RA. Reliability of classifying multiple sclerosis disease activity using magnetic resonance imaging in a multiple sclerosis clinic. *JAMA Neurol.* 2013;70(3):338–344.
169. Narayana PA, Zhou Y, Hasan KM, Datta S, Sun X, Wolinsky JS. Hypoperfusion and T1-hypointense lesions in white matter in multiple sclerosis. *Mult Scler.* 2014;20(3):365–373.
170. Akaishi T, Nakashima I, Mugikura S, Aoki M, Fujihara K. Whole brain and grey matter volume of Japanese patients with multiple sclerosis. *J Neuroimmunol.* 2017;306:68–75.
171. Lazeron RH, Boringa JB, Schouten M, et al. Brain atrophy and lesion load as explaining parameters for cognitive impairment in multiple sclerosis. *Mult Scler.* 2005;11(5):524–531.
172. Hannoun S, Koccar G, Codjia P, et al. T1/T2 ratio: a quantitative sensitive marker of brain tissue integrity in multiple sclerosis. *J Neuroimaging.* 2022;32(2):328–336.
173. Boaventura M, Sastre-Garriga J, Garcia-Vidal A, et al. T1/T2-weighted ratio in multiple sclerosis: a longitudinal study with clinical associations. *Neuroimage Clin.* 2022;34: 102967.
174. Coll L, Pareto D, Carbonell-Mirabent P, et al. Deciphering multiple sclerosis disability with deep learning attention maps on clinical MRI. *Neuroimage Clin.* 2023;38: 103376.
175. Hosseinpour Z, Jonkman L, Oladosu O, et al. Texture analysis in brain T2 and diffusion MRI differentiates histology-verified grey and white matter pathology types in multiple sclerosis. *J Neurosci Methods.* 2022;379: 109671.